Class Imbalance in Histopathology: Skewed Prior and Thin Tail Support Compound on TCGA-UT

Yannik Queisler

September 22, 2026

Abstract

On TCGA-UT, a long-tailed training allocation at fixed patients and fixed patch budget costs 2.76 points of balanced accuracy at an imbalance ratio of $\rho = 100$. Each ratio changes two things at once: the class prior of the training objective, and the number of patches that represent each tail class. This report separates the two. For every ρ from 2 to 100, a *prior-only* arm trains on the balanced rows with class weights that impose the long-tailed prior, and a *support-only* arm trains on the long-tailed rows with class weights that restore a balanced prior. All arms use the same patients, patches, and class order as the original curve, so every contrast is paired. At $\rho = 100$ the prior alone costs 1.04 points (95% interval: 0.69 to 1.38) and the thin support alone 0.54 points (0.37 to 0.71). Their sum falls short of the full damage by 1.18 points (1.02 to 1.34), so the two factors interact: the skewed prior hurts much more when the tail is also thinly sampled. Up to $\rho = 20$ the prior carries most of the damage, and support alone costs nothing. The acceleration of the curve beyond $\rho = 10$ comes from support and from the interaction; the prior-only curve stays linear in $\log \rho$. Subtracting the log prior from the logits recovers half of the full damage at $\rho = 100$ but falls 0.73 points short of the support-only arm, and it lowers the accuracy of the prior-only arm. The skewed prior also drives the drop in selected regularization and the rise in fitted temperature that accompany imbalance.

1 Motivation

The prevalence curve of this series measured how balanced accuracy on TCGA-UT degrades as the training class distribution moves from balanced to long-tailed, with twenty patients per class and a total of 19,200 patches held fixed (Queisler, 2026d). The damage was 0.57 points at $\rho = 10$ and 2.76 points at $\rho = 100$, and the loss per doubling of ρ rose from 0.17 points below $\rho = 10$ to 0.65 points above it. Because the budget is fixed, each ratio moves two factors together. The *class prior*, the share of the training objective that each class contributes, tilts toward the head. The *tail support*, the number of patches from which the classifier estimates each rare class, shrinks; at $\rho = 100$ the smallest class keeps 28 patches, between one and two per patient. The curve could not say which factor causes the damage, and the companion BRACS study proposed, without testing it, that a shifted prior rather than thin support explains its much larger damage (Queisler, 2026c).

This report separates the two factors on TCGA-UT. The question is: *does the prevalence damage come from the skewed prior, from the thin tail support, or from their interaction, and which of them makes the curve accelerate beyond $\rho = 10$?*

2 Design

The study reuses the full design of the prevalence curve (Queisler, 2026d): the 30 TCGA-UT cancer types with frozen Virchow2 features (Zimmermann et al., 2024), three patient-disjoint splits with ten draws each, twenty patients per class, a total budget of $T = 19{,}200$ training

patches, the exponential long-tail profile with a random class order per draw, and nested patch orderings within each patient. The balanced arm r1 gives each class 640 patches; the ratio arm rρ gives the class at rank k a count proportional to $\rho^{-k/29}$. Both are taken unchanged from the stored fits of that study.

Each arm is now described by two allocations: the *data allocation*, which rows enter training, and the *prior allocation*, which class shares the loss is weighted toward. With n_c training rows of class c out of N in the data allocation and t_c rows out of T in the prior allocation, every row of class c receives the weight

$$w_c = \frac{t_c/T}{n_c/N}. \tag{1}$$

The weights sum to N, so the regularization strength keeps the same scale as in the unweighted fits, and the weighted loss gives class c exactly the share t_c/T. Two arms are added per ratio (Table 1). The *prior-only* arm Pρ trains on the rows of r1 and weights them toward the class shares of rρ. Its objective has the same expectation as that of rρ, but every class, including the tail, is estimated from 640 patches. The *support-only* arm Sρ trains on the rows of rρ and weights them back to balanced shares, so it sees the thin tail of rρ under a balanced objective. At $\rho = 100$, P gives the head class a weight of 4.44 and the tail class 0.044, and S gives the head 0.23 and the tail 22.9.

Arm	Training rows	Prior of the loss	Source
r1	balanced, 640 per class	balanced	prevalence curve
rρ	long-tailed, ratio ρ	long-tailed (implicit)	prevalence curve
Pρ	balanced, 640 per class	long-tailed, via weights	fitted here
Sρ	long-tailed, ratio ρ	balanced, via weights	fitted here
LPρ, Lrρ	as Pρ, rρ	logit-adjusted	no fit

Table 1: The four fitted arms form a two-by-two design per ratio, crossing the data allocation with the prior of the loss. The prior-only arm Pρ and the support-only arm Sρ each change one factor relative to the balanced arm; the ratio arm rρ changes both. The logit-adjusted arms reuse the stored test probabilities of Pρ and rρ. All arms share patients, patch orderings, and class order per split and draw.

Two further arms need no fit. They apply post-hoc *logit adjustment* (Menon et al., 2021) to the uncalibrated test probabilities of Pρ and rρ, predicting $\arg\max_c (\log p_c - \log \pi_c)$ with π_c the training prior share of class c. They show how much of each arm's damage a correction of the class intercepts alone removes.

Every fitted arm selects its regularization strength from nine values between 10^{-6} and 100 by validation patient-macro balanced accuracy (Queisler, 2026b) and then fits one validation temperature (Guo et al., 2017), as in the prevalence curve. The ratios are $\rho \in \{2, 5, 10, 20, 50, 100\}$. The study fits 2 arms $\times$ 6 ratios $\times$ 30 split-draw pairs, 360 classifiers, and all of them converged. The reused r arms reproduce the published damages exactly.

3 Evaluation

The endpoint is the damage relative to the balanced arm, $\Delta_A(\rho) = \mathrm{BA}(\mathrm{r1}) - \mathrm{BA}(A\rho)$, for $A \in \{\mathrm{r}, \mathrm{P}, \mathrm{S}\}$, where BA is test patient-macro balanced accuracy. The *interaction* is the part of the full damage that the two single-factor damages do not add up to,

$$I(\rho) = \Delta_{\mathrm{r}}(\rho) - \Delta_{\mathrm{P}}(\rho) - \Delta_{\mathrm{S}}(\rho), \tag{2}$$

and the *shares* $\Delta_{\mathrm{P}}/\Delta_{\mathrm{r}}$ and $\Delta_{\mathrm{S}}/\Delta_{\mathrm{r}}$ express the single-factor damages as fractions of the full damage. Shares are reported only where $\Delta_{\mathrm{r}} \geq 1$ point, that is from $\rho = 50$ on; the primary ratio is $\rho = 100$. *Acceleration* is the least-squares slope of the damage on $\log_2 \rho$ over $\rho \geq 10$ minus the slope over $\rho \leq 10$, per family, in points per doubling. For the r family the low segment spans $\rho = 1$ to 10, as in the prevalence curve; for P and S it spans $\rho = 2$ to 10.

The first secondary endpoint is the fraction of the prior damage that logit adjustment recovers, the *intercept share*

$$\frac{\mathrm{BA}(\mathrm{LP}\rho) - \mathrm{BA}(\mathrm{P}\rho)}{\Delta_{\mathrm{P}}(\rho)}. \tag{3}$$

The others are the difference $\mathrm{BA}(\mathrm{Lr}\rho) - \mathrm{BA}(\mathrm{S}\rho)$ between post-hoc and training-time correction of the same data; the recall of the head, body, and tail thirds of the classes by assigned rank; and the selected regularization strength and mean fitted temperature per arm. Pooling and uncertainty follow the prevalence curve: equal weight per split, Bayesian bootstrap weights on test patients (Rubin, 1981) combined with resampling of draws within splits, 2,000 shared replicates, and 95 percent percentile intervals of every paired contrast.

Four readings were fixed in advance, at the primary ratio. The *prior carries* the damage if $\Delta_{\mathrm{P}} \geq 1$ point with a share of at least one half while $\Delta_{\mathrm{S}} < 1$ point; *support carries* it under the mirror conditions. The factors are *additive* if the interval of I contains zero, and they *interact* if I is resolved above zero. The acceleration is attributed to the family whose slope difference is resolved above zero.

4 Results

The results follow the endpoints: first the decomposition of the damage per ratio, then the acceleration, the classes on which each factor's damage lands, the effect of logit adjustment, and finally the selected regularization and temperatures. The balanced arm reaches 72.05 percent balanced accuracy (70.68 to 73.31), as in the prevalence curve.

4.1 Decomposition of the damage

The prior alone accounts for most of the damage at moderate ratios, the thin support alone for none of it, and at large ratios the two factors together cost far more than their sum (Table 2, Figure 1). Up to $\rho = 20$ the prior-only damage follows the full damage closely: 0.49 against 0.57 points at $\rho = 10$, 0.71 against 0.996 points at $\rho = 20$. The support-only arm stays within 0.1 points of the balanced arm over the same range, and its intervals contain zero from $\rho = 2$ to 20. A balanced objective on the long-tailed rows therefore loses nothing as long as the tail class keeps at least five patches per patient.

Beyond $\rho = 20$ the picture changes. Support alone starts to cost accuracy, 0.18 points at $\rho = 50$ and 0.54 points at $\rho = 100$, when the tail class keeps between one and three patches per patient. The prior-only damage keeps growing slowly and reaches 1.04 points at $\rho = 100$. The interaction is resolved from $\rho = 10$ on and roughly doubles with each step of the grid, from 0.14 points at $\rho = 10$ to 1.18 points at $\rho = 100$, where it exceeds the prior-only damage. At the primary ratio the prior accounts for 38 percent of the full damage, support for 19 percent, and the interaction for the remaining 43 percent. The three splits agree: the interaction at $\rho = 100$ is 1.29, 1.11, and 1.14 points.

4.2 Acceleration

The three families accelerate differently (Table 3). The full damage grows by 0.17 points per doubling of ρ up to $\rho = 10$ and by 0.65 points beyond, an acceleration of 0.48 points (0.40 to 0.56). The prior-only damage is linear in $\log \rho$: 0.17 points per doubling below $\rho = 10$ and 0.16

ρ	Δ_r (both)	Δ_P (prior)	Δ_S (support)	Interaction I	Share P	Share S
2	0.13 [0.03, 0.23]	0.10 [0.00, 0.20]	0.01 [−0.02, 0.04]	0.02 [−0.02, 0.07]	—	—
5	0.36 [0.13, 0.57]	0.33 [0.12, 0.54]	0.00 [−0.05, 0.05]	0.02 [−0.04, 0.08]	—	—
10	0.57 [0.30, 0.84]	0.49 [0.25, 0.72]	−0.05 [−0.12, 0.02]	0.14 [0.07, 0.21]	—	—
20	0.996 [0.66, 1.34]	0.71 [0.43, 0.98]	−0.06 [−0.15, 0.02]	0.35 [0.25, 0.46]	—	—
50	1.80 [1.41, 2.19]	0.90 [0.59, 1.21]	0.18 [0.04, 0.32]	0.72 [0.59, 0.85]	0.50	0.10
100	2.76 [2.34, 3.20]	1.04 [0.69, 1.38]	0.54 [0.37, 0.71]	**1.18** [1.02, 1.34]	0.38	0.19

Table 2: The skewed prior carries most of the damage up to $\rho = 20$; beyond it the interaction of prior and thin support grows fastest. Damages are balanced-accuracy losses against the balanced arm in percentage points, with paired 95% bootstrap intervals; the interaction is $\Delta_r - \Delta_P - \Delta_S$. Shares are single-factor damages as fractions of Δ_r and are omitted where $\Delta_r < 1$ point. Their intervals at $\rho = 100$ are 0.29 to 0.45 for P and 0.14 to 0.25 for S.

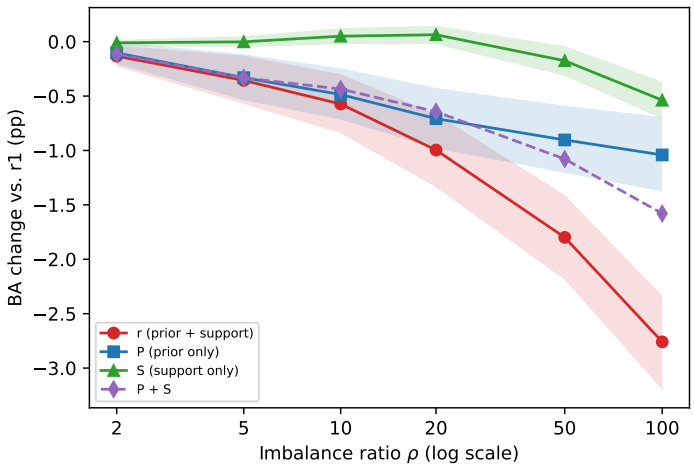


Figure 1: The full damage exceeds the sum of its single-factor parts at every ratio from $\rho = 10$ on. Change in patient-macro balanced accuracy relative to the balanced arm for the ratio arm r (prior and support), the prior-only arm P, and the support-only arm S, each with a shaded paired 95% bootstrap band; the dashed line is the sum of the P and S changes. All series are plotted at the realized ratio of rρ.

above, a difference whose interval, −0.06 to 0.06, is centred on zero. The support-only damage is flat below $\rho = 10$ and grows by 0.18 points per doubling above it, a resolved acceleration of 0.20 points (0.16 to 0.24). The single factors together account for 0.20 of the 0.48 points of acceleration; the rest comes from the growing interaction.

Family	Slope, $\rho \leq 10$	Slope, $\rho \geq 10$	Acceleration
r (both)	0.17 [0.08, 0.26]	0.65 [0.59, 0.71]	**0.48** [0.40, 0.56]
P (prior)	0.17 [0.08, 0.24]	0.16 [0.12, 0.20]	−0.00 [−0.06, 0.06]
S (support)	−0.03 [−0.05, 0.00]	0.18 [0.14, 0.22]	**0.20** [0.16, 0.24]

Table 3: The prior-only damage grows at a constant rate per doubling, while the support-only damage appears only beyond $\rho = 10$. Least-squares slopes of the damage on $\log_2 \rho$ in points per doubling, with paired 95% bootstrap intervals; acceleration is the high-segment slope minus the low-segment slope, in bold where resolved above zero. The low segment of r includes the balanced arm at $\rho = 1$; those of P and S start at $\rho = 2$.

4.3 Where the damage lands

Both factors move recall from the tail third of the classes to the head third, but they do so in different proportions, and neither reaches the redistribution of the ratio arm (Table 4). At $\rho = 100$ the prior-only arm lowers tail recall by 4.9 points and raises head recall by 2.6 points, and the support-only arm lowers the tail by 3.7 and raises the head by 2.1, although its objective is balanced. The ratio arm lowers the tail by 11.5 points, 2.8 points more than the two single-factor losses together, and it also lowers the body by 2.1 points, where the single-factor arms lose 0.8 points or nothing. The interaction thus falls mainly on the tail and body; the head gains of the ratio arm exceed the single-factor sum by only 0.7 points. Below $\rho = 20$ the support-only arm leaves all three thirds within one point of the balanced arm, whereas the prior-only arm already shifts 4.0 points of tail recall at $\rho = 10$.

		$\rho = 10$			$\rho = 20$			$\rho = 100$		
Third	r1	r	P	S	r	P	S	r	P	S
Head	72.04	76.09	74.86	72.74	76.60	74.92	73.13	77.38	74.65	74.09
Body	72.18	71.46	71.88	72.12	71.01	71.69	72.04	70.05	71.38	72.26
Tail	71.93	66.90	67.97	71.46	65.56	67.42	71.18	60.46	67.01	68.19

Table 4: The combination of prior and thin support costs the tail more than the two factors separately. Mean patient-macro recall in percent of the ten head, body, and tail classes of each draw, by their assigned rank, averaged over 30 fits. The r columns are taken from the prevalence curve.

4.4 Logit adjustment

Post-hoc logit adjustment repairs part of the ratio arm's damage but none of the prior-only arm's (Table 5). Applied to r100, it raises balanced accuracy from 69.29 to 70.79 percent and removes about half of the 2.76-point damage. It still falls 0.73 points (0.40 to 1.04) short of the support-only arm, which trains on the same rows with a balanced objective, and it falls short of it at every ratio from $\rho = 5$ on. Applied to Pρ, the same correction lowers balanced accuracy, by 0.57 points at $\rho = 100$. The intercept share of the prior damage is therefore negative: at $\rho = 50$ and $\rho = 100$ the adjustment adds about half to the prior-only damage (54 percent at $\rho = 100$,

23 to 109) instead of removing part of it, and at $\rho \leq 20$ its effect is not resolved. Subtracting the full log prior thus overcorrects the prior-only arm. This indicates that the weighted fit has not shifted its class intercepts by the full log prior; its damage lies in how the class weights change the fitted scores, not in a uniform intercept offset.

ρ	r	Lr	P	LP	S	Lr − S
10	71.48	71.80	71.57	71.54	72.10	−0.31 [−0.51, −0.10]
20	71.06	71.56	71.35	71.21	72.12	−0.56 [−0.81, −0.30]
50	70.25	71.20	71.15	70.74	71.88	−0.68 [−0.98, −0.38]
100	69.29	70.79	71.01	70.45	71.52	−0.73 [−1.04, −0.40]

Table 5: Logit adjustment helps the ratio arm but stays below training-time rebalancing, and it hurts the prior-only arm. Test patient-macro balanced accuracy in percent; the balanced arm reaches 72.05. Lr and LP subtract the log training prior from the logits of r and P; the last column is the paired difference between post-hoc and training-time correction of the same rows, with its 95% bootstrap interval.

4.5 Regularization and temperature

The prevalence curve found that stronger imbalance selects weaker regularization and needs a larger temperature, and it could not say whether the prior or the support causes this (Queisler, 2026d). The prior does. The support-only arms select $\lambda = 0.1$ in 26 to 28 of 30 fits and $\lambda = 0.01$ in the rest at every ratio, like the balanced arm, and their mean temperature stays between 0.85 and 0.87 from $\rho = 2$ to 100, against 0.85 for r1. The prior-only arms move toward weaker regularization as ρ grows: $\lambda = 0.1$ in 20 of 30 fits at $\rho = 2$, $\lambda = 0.01$ in 25 of 30 at $\rho = 10$, and $\lambda \leq 10^{-3}$ in all 30 at $\rho = 100$, the same choice as r100. Their mean temperature rises from 0.94 at $\rho = 2$ to 1.85 at $\rho = 100$, below the 2.16 of r100. One prior-only fit at $\rho = 100$ selects the smallest grid value, 10^{-6}. Refitted with λ extended down to 10^{-10}, it keeps this value, with 71.27 percent validation balanced accuracy against at most 71.26 percent below it, so the grid boundary does not constrain the selection.

5 Discussion

Of the readings fixed in Section 3, the results select the interaction reading. At $\rho = 100$ the prior-only damage exceeds one point but carries only 38 percent of the full damage, the support-only damage stays below one point, and the interaction is resolved at 1.18 points. At $\rho = 20$, where the full damage of 0.996 points falls just short of one point, the interaction of 0.35 points is already resolved, and the prior-only damage of 0.71 points stays below the one-point condition. The acceleration is attributed to support, the only single factor with a resolved slope difference, but the interaction contributes more than half of it.

The two factors thus play different roles along the curve. At moderate ratios, the prevalence damage on TCGA-UT is a prior effect: a balanced objective on the long-tailed rows loses nothing up to $\rho = 20$, so five patches per patient are enough to estimate a tail class once its share of the loss is restored. The prior alone produces a steady loss of about 0.17 points per doubling of ρ, together with the shift toward weaker regularization and higher temperature. At large ratios the two factors compound. A skewed prior costs 1.04 points when every class keeps 640 patches, but 1.04 plus 1.18 points when the tail is also thin. A plausible mechanism is that a tail class estimated from one or two patches per patient has noisy and weak scores, so the head-ward push of the prior moves more of its test patches across the boundary than it does for a well-estimated

class. The tail and body losses of Table 4, which exceed the single-factor sum by more than the head gains do, fit this account. The study does not test the mechanism directly.

For mitigation, the decomposition points to training-time rebalancing. On the rows of rρ, restoring a balanced objective by class weights removes the prior damage and the interaction together and leaves only the support-only damage, 0.54 points at $\rho = 100$. Post-hoc logit adjustment of the same classifier recovers only half of the damage and stays 0.73 points below that level, and it does not help a classifier whose prior was imposed by weights. On TCGA-UT, the cheap post-hoc correction is therefore not a substitute for rebalancing during training. The support-only damage that remains is small compared with the patient-count effects of this series (Queisler, 2026a).

The companion BRACS study suggested that its large damage comes from a shifted prior under class overlap, not from thin tail support (Queisler, 2026c). On TCGA-UT, the prior alone at full support costs at most 1.04 points, far below the 7.98 points of BRACS at $\rho = 100$, and the interaction with thin support is larger than the prior effect itself. The BRACS explanation thus needs its own test, with the same design on BRACS, before the prior can be credited with the BRACS damage.

6 Limitations

The prior-only arm imposes the long-tailed prior by class weights, not by resampling. Its weighted objective has the same expectation as that of the ratio arm, but it is not the same estimator: the head rows of P number 640 rather than up to 2,843, and weighting and sampling can reach different solutions for a regularized classifier, as the adverse effect of logit adjustment on P indicates. The interaction therefore mixes the effect of thin support under a skewed prior with any difference between weighting and sampling. A design that resamples the head at full tail support would separate them, but at a fixed patient depth of 160 patches it is feasible only up to $\rho \approx 5$. Each fitted arm selects its own regularization, so the decomposition includes the effect of the prior on model selection; with λ held fixed, the prior-only damage could differ.

The acceleration segments of the P and S families start at $\rho = 2$, whereas that of the r family starts at $\rho = 1$; the low-segment slopes of P and S are therefore estimated from three points. The full damage at $\rho = 20$ falls just short of the one-point threshold, so shares are reported only from $\rho = 50$ on. All results are conditional on twenty patients per class, frozen Virchow2 features, multinomial logistic regression, TCGA-UT, and three overlapping splits.

References

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330. PMLR, 2017.

Aditya Krishna Menon, Sadeep Jayasumana, Ankit Singh Rawat, Himanshu Jain, Andreas Veit, and Sanjiv Kumar. Long-tail learning via logit adjustment. In *International Conference on Learning Representations*, 2021. URL `https://openreview.net/forum?id=37nvvqkCo5`.

Yannik Queisler. Patient shortage in histopathology: Class-centre error explains most of the benefit of additional patients. Technical report, TU Berlin, 2026a. Companion centre-error report, Experiment 16.

Yannik Queisler. Patient shortage in histopathology: Classifier or coverage limitation? Technical report, TU Berlin, 2026b. Companion patient-shortage report.

Yannik Queisler. Class imbalance in histopathology: Prevalence alone costs several points on BRACS. Technical report, TU Berlin, 2026c. Companion BRACS prevalence report, Experiment 26.

Yannik Queisler. Class imbalance in histopathology: Prevalence alone costs little accuracy on TCGA-UT. Technical report, TU Berlin, 2026d. Companion TCGA-UT prevalence report, Experiment 25.

Donald B. Rubin. The Bayesian bootstrap. *The Annals of Statistics*, 9(1):130–134, 1981. doi: 10.1214/aos/1176345338.

Eric Zimmermann, Eugene Vorontsov, Julian Viret, Adam Casson, Michal Zelechowski, George Shaikovski, Neil Tenenholtz, James Hall, Thomas Fuchs, Nicolo Fusi, Siqi Liu, and Kristen Severson. Virchow2: Scaling self-supervised mixed magnification models in pathology. *arXiv preprint arXiv:2408.00738*, 2024. doi: 10.48550/arXiv.2408.00738.